

Building the CSA's Artificial Intelligence Capability

A Business Case for a Secure, Shared AI Experimentation Environment

Executive Summary:

Artificial intelligence could significantly improve scientific, engineering and corporate work at the CSA. However, the Agency currently lacks a secure and accessible environment in which to evaluate these capabilities. This horizontal initiative proposes testing CRAFT through SSC's GC Cloud One: LaunchPad using controlled use cases and measurable performance indicators. The results will support a decision to discontinue, refine or advance each capability toward an operational pilot.

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1. INTRODUCTION

1.1. The Transformative Potential of Artificial Intelligence

Artificial intelligence has the potential to improve how information is collected, structured, analyzed and communicated. It can process large volumes of structured and unstructured information, identify patterns and inconsistencies, automate repetitive activities, and make specialized knowledge more accessible to employees.

When combined with authoritative documents, databases and appropriate business rules, AI can support scientists, engineers and analysts throughout the information lifecycle. Potential applications include extracting information from technical documents, organizing data in structured repositories, searching across multiple information sources, generating analyses and visualizations, identifying anomalies, and preparing draft reports and recommendations.

AI can also improve decision support by helping users bring together information that is currently dispersed across documents, spreadsheets, databases and corporate systems. This could reduce the time spent locating and preparing information, allowing specialists to focus more attention on interpretation, validation and higher-value work.

The objective is not to replace scientific, engineering or financial judgment. Rather, AI can augment professional expertise by accelerating routine tasks, highlighting issues that warrant attention and presenting relevant evidence to support human decisions. To realize these benefits responsibly, the CSA must develop the capacity to evaluate AI technologies, understand their limitations and determine where they can provide measurable operational value.

1.2. Problem Statement

The CSA currently lacks an approved, secure and accessible experimentation environment where scientists, engineers and analysts can develop and test new technologies, data solutions and innovative digital capabilities. Access to essential tools -including artificial intelligence, cloud computing, software-development platforms, data-processing services, application hosting and SQL databases - is limited, fragmented or unavailable to users.

Existing corporate environments are largely siloed, with information distributed across disconnected systems, documents and spreadsheets, and without a common data model or authoritative single source of truth.

Existing corporate environments are primarily designed to support stable operational systems. They are not necessarily configured to support rapid experimentation, iterative development or the temporary deployment of proofs of concept. Obtaining access to new tools and services can involve lengthy processes, while locally developed solutions may not have an approved pathway for testing, integration or eventual transition to production.

These constraints make it difficult to explore emerging technologies, integrate and analyze complex datasets, develop prototypes, conduct proofs of concept, and evaluate potential solutions before committing to operational implementation. They also limit collaboration

across business areas and reduce opportunities for scientists, engineers, analysts and digital specialists to work together on shared technical challenges.

The absence of an experimentation environment also restricts the CSA's ability to build internal knowledge and technical capacity. Without practical experience, the Agency cannot easily assess the accuracy, limitations, security implications, costs or operational value of emerging technologies. This makes it more difficult to define informed requirements, challenge vendor claims, select appropriate solutions and oversee externally delivered work.

As AI and cloud technologies are adopted more broadly across government, the CSA risks falling behind organizations that already provide their employees with controlled opportunities to experiment and learn. The challenge is therefore broader than access to individual tools: the Agency would benefit from an enabling environment in which innovation can be explored safely, evidence can be collected, and promising concepts can be evaluated before investments or operational commitments are made.

1.3. Merits of Controlled Experimentation

Controlled experimentation provides a low-risk, evidence-based way to explore emerging technologies and determine whether they can solve real business problems. Instead of committing immediately to a large implementation, the CSA can begin with bounded use cases, synthetic or approved data, limited system access and clearly defined success criteria.

An experimentation approach allows the Agency to test both the technology and the conditions required for its successful adoption. It can help determine:

- whether the proposed capability is technically feasible;
- whether it produces accurate, reliable and traceable results;
- whether it can be integrated with existing documents, databases and workflows;
- what security, privacy and information-management controls are required;
- whether users find the capability useful and trustworthy;
- what skills, governance and operational support would be needed; and
- whether the expected benefits justify the implementation and operating costs.

Experimentation also provides value when a concept does not succeed. Identifying technical limitations, poor-quality data, unsuitable use cases or excessive costs at an early stage prevents larger investments in solutions that may not meet the Agency's needs. The result is a more informed basis for deciding whether to discontinue, refine, scale or operationalize a capability.

1.4. About GC Cloud One

Shared Services Canada (SSC) recently introduced **GC Cloud One: LaunchPad** as part of its initiative to progressively centralize cloud services across the Government of Canada. LaunchPad was created, in part, to address the need for controlled cloud environments where federal scientists and engineers can conduct research, experimentation and proofs of concept.

LaunchPad can provide a cloud-based experimentation environment capable of handling information up to **Protected B**. The proposed environment may include Amazon Web Services (AWS), artificial intelligence capabilities, data-processing and storage services, and a SQL database for structuring and retaining experimental data.

2. PROPOSED EXPERIMENT

A horizontal initiative is proposed across CSA branches and directorates to develop and evaluate **CRAFT - Collaborative Reasoning and Federated Technical Knowledge** - as a shared AI experimentation platform.

CRAFT would support the controlled exploration of AI applications across scientific, engineering, operational and corporate activities. Potential use cases include document and knowledge management, risk assessment, cost estimation, anomaly detection, mission-planning support, data analysis, and report generation.

2.1. Experimentation objectives

The experiment will evaluate a progressive set of AI-enabled capabilities, beginning with low-risk, controlled use cases. These may include extracting information from test documents, proposing values for structured SQL fields, answering questions using documents and database records, generating tables and charts, drafting analytical observations, guiding users through the costing process, and supporting data-quality reviews and imputation.

The initial experiment will use synthetic, anonymized or otherwise approved information.

The experiment is intended to answer three fundamental questions:

1. Can the AI produce sufficiently accurate, consistent and traceable results?
2. Can it be integrated safely with documents, SQL databases and analytical tools?
3. Does it create measurable value compared with the current way of working?

The results will inform the organization's AI Capability and Assurance Maturity Model and help determine which capabilities are suitable for further development, operational deployment or discontinuation.

The experiment will conclude with a final report documenting the findings, lessons learned, performance against the established success criteria, and recommendations on whether each capability should be discontinued, further developed or advanced to a controlled operational pilot.

2.2. Multidisciplinary approach towards a common solution

The experiment would be conducted as a horizontal initiative involving several teams across the CSA, with each participating area contributing its expertise and evaluating use cases relevant to its mandate:

- **Finance:** AI-assisted cost estimation, accounting, financial analysis and strategic planning, including document analysis, data-quality review, anomaly detection and report generation.
- **Space Utilization:** AI-enabled access to, processing and analysis of Earth observation data, including data-cube applications, automated information extraction, pattern detection and decision support.
- **Space Science and Technology:** AI applications for the support of mission operations, including mission planning, telemetry analysis, anomaly detection, operational decision support and the generation of mission reports.

Although the use cases differ, the participating teams would share a common experimentation environment, technical architecture and governance framework. This horizontal approach would encourage collaboration, reduce duplication and generate reusable knowledge and components for the Agency.

3. MEASURING SUCCESS

Maturity models provide a structured method for assessing current capabilities and defining a progressive path toward more advanced, reliable and controlled practices. The AI Capability and Assurance Maturity Model will be used to evaluate each use case consistently and determine whether it is ready to progress from experimentation to controlled operational deployment.

In addition to the maturity model, each use case will be evaluated using measurable key performance indicators. These indicators may include response time, extraction and classification accuracy, error and hallucination rates, the accuracy of information retrieved and federated across multiple sources, source traceability, consistency of repeated results, human correction rates, system availability, user satisfaction, time saved and operating cost.

Performance thresholds will reflect the risk and intended use of each capability. Higher levels of accuracy, traceability and human validation will be required for database transactions, financial analysis and operational decision support than for lower-risk activities such as summarization or drafting.

3.1. AI Capability and Assurance Maturity Model

Level	Maturity	Description	Example capabilities
0	Manual	Work is completed manually using documents, spreadsheets, SAP and SQL.	Manual data entry, searches, calculations and reporting.
1	AI-Assisted	AI helps users create, summarize or classify information but does not access authoritative systems.	Summarize documents, draft reports, explain costing methods, generate SQL queries.

2	Grounded AI	AI retrieves information from approved documents and databases and identifies its sources.	Answer questions using documents and SQL; extract fields from test documents; provide citations and confidence scores.
3	Human-Validated Automation	AI recommends structured data or actions, but a person must approve them.	Extract document data, map it to SQL fields, flag uncertainties, and submit only after human confirmation.
4	Controlled AI Agent	AI performs approved, repeatable actions within defined limits and maintains an audit trail.	Populate tables, run analyses, generate charts, clean data, identify SAP anomalies and prepare draft reports.
5	Adaptive Decision Support	AI coordinates several capabilities, learns from validated outcomes and proactively identifies issues.	Detect cost trends, recommend imputations, explain anomalies, propose corrective actions and guide the complete costing process.

3.2. Maturity Assessment Dimensions

Each use case could be scored from 0 to 5 against the following dimensions:

1. Data readiness
2. AI capability
3. System integration
4. Human oversight
5. Reliability and validation
6. Explainability and traceability
7. Security and privacy
8. Governance and accountability
9. Operational adoption
10. Performance monitoring

This prevents the model from treating technical sophistication as the only measure of maturity. A chatbot may be technically advanced but still immature if it has weak data, no citations, no validation and no audit trail.

3.3. Capability Streams under consideration

Capability stream	Initial capability	More mature capability
Document intelligence	Summarize documents	Extract, classify and map information into SQL
Knowledge retrieval	Search documents	Answer questions across documents and structured data with citations

Data interaction	Generate SQL	Execute controlled queries and approved POST operations
Analytics	Populate tables and charts	Select methods, run analyses and explain results
Reporting	Draft narrative text	Generate traceable reports with findings, limitations and recommendations
Process guidance	Answer costing questions	Guide users through the appropriate workflow based on project context
Data quality	Identify obvious errors	Detect anomalies, duplicates and inconsistencies across SAP and other systems
Imputation	Suggest missing values	Recommend defensible imputation methods with uncertainty and pedigree
Decision support	Present results	Identify implications, risks and possible actions
Governance	Informal human review	Role-based approvals, audit logs, monitoring and formal accountability

4. PROGRAM TERMS AND CONDITIONS

4.1. Budget

The total expected budget for the experiment is 100K covering 12 calendar months.

4.2. Procurement strategy

Services would be prepaid within a Gov't of Canada Cloud Framework Agreement. Funds would be transferred via a journal entry.

As services are consumed, the associated amount is deducted against the prepaid amount on a monthly basis. Shared Services would charge a 10% brokering fee as coordinator of the services. Training Advisory and Support services are also available.

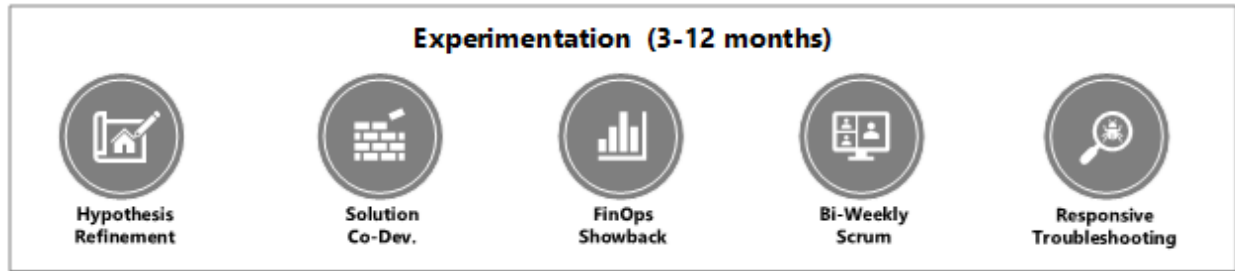
Once the experimentation is completed, if the CSA decides to expand or continue into operations, then a cost estimate would be required.

The procurement strategy changes depending on the value of the project:

- GC organizations can use LaunchPad for a single experiment valued up to \$500,000 (+taxes).
- If the total project cost is between 500K and 4.5M, Shared Services will issue a standing offer
- If the project total is higher than 4.5M, it must be competed.

4.3. Schedule and Life-Cycle

The expected duration of experimentation is 12 months.



4.4. Security Considerations

4.4.1.1. Security classification

Although the services is rated Protected B, the experiment will begin using unclassified data. Fake (dummy) data will be produced for testing purposes. As the proof-of-concept advances, it may be necessary to add protected B data. Where possible, we would anonymize, modify and obfuscate the data to protect sensitive info.

For example: if the focal plane array resolution is protected B, we could replace the number with a different one (a fake one).

4.4.1.2. Connectivity

The cloud environment remains separate from the CSA internal network. There is no persistent network connection between the cloud environment and GC systems.

Authorized users or GC system may send an approved copy of data from inside the GC network to the cloud environment through an approved transfer mechanism.

In practical terms, the experimentation environment will permit:

- Export approved data from SAP, SQL, Excel or a document repository and upload or push the copy to LaunchPad.
- Periodically send an approved dataset through an authorized HTTPS endpoint, API or file-transfer service, if permitted by the applicable security rules.

If the experimentation eventually leads to a production environment, at a different service level, then we can examine the pros and cons of:

- Letting LaunchPad connect directly into the CSA network to query SAP, Oracle, shared drives or internal repositories.
- Establishing a VPN or SCED connection that makes LaunchPad part of, or privately connected to, the GC network.

4.4.1.3. Human Control and Accountability

CRAFT is designed to assist people, not replace them. Human authority remains central, especially for important decisions.

The system requires human review for high-risk actions. These include cost commitments, updates to controlled risk records, and mission parameter changes. Reviewers must be able to approve, reject, or modify recommendations before they are committed.

When a recommendation is sent for review, CRAFT must show the supporting evidence, the reasoning path, and the confidence score. This helps the reviewer make an informed decision.

This approach supports accountability. It keeps people responsible for final decisions while allowing the system to reduce effort, organize evidence, and improve consistency.

4.5. Governance and Authorities

The Cloud Framework Agreement with shared services requires the signature of a director or higher.

Governance would require consultation with different stakeholders, and various approval boards.

Appendix A - Mapping Costing Use Cases

A.1. Intelligent Document Intake

AI reads documents and:

- identifies the document type;
- extracts candidate values;
- maps them to SQL tables and fields;
- shows the source passage;
- assigns a confidence level;
- highlights ambiguous or missing information;
- presents the proposed values to a person;
- performs the POST only after approval.

Capability Level 3 because it demonstrates useful automation without allowing uncontrolled database changes.

A.2. Conversational knowledge retrieval

AI combines information from:

- policies, standards and project documents;
- historical estimates;
- the costing semantic model;
- SAP or financial extracts;
- schedules and engineering parameters.

It answers in plain language while identifying the source, reporting date and applicable baseline. This progresses from Capability Level 2 retrieval to Level 4 when it can safely execute approved queries.

A.3. Conversational analytics

A user could ask: Show annual expenditures by mission phase and compare actual costs against the approved baseline.

The AI would:

1. interpret the request;
2. generate the SQL;
3. validate the fields and filters;
4. execute the query;
5. populate a table or chart;
6. describe assumptions and limitations.

A.4. Analytical interpretation and reporting

AI interprets results and drafts:

- key findings;
- trends and anomalies;
- cost and schedule variances;
- uncertainty explanations;
- management implications;
- recommended follow-up analysis.

At a mature level, every statement should be traceable to a calculation, query or source—not merely plausible AI-generated commentary.

A.5. Costing process advisor

The chatbot determines what the user is trying to cost and guides them through:

- selecting the appropriate costing approach;
- determining the required CBS level;
- identifying data requirements;
- choosing estimation methods;
- performing risk and uncertainty analysis;
- conducting cross-checks;
- preparing documentation;
- completing the Cost Review and executive presentation.

This could eventually adapt its guidance based on project value, phase, CRL, OPMCA, available data and required level of assurance.

A.6. Data quality and imputation

AI could:

- detect inconsistent SAP coding;
- identify duplicate or incorrectly classified transactions;
- find abnormal labour or procurement costs;
- reconcile SAP data with project records;
- detect missing engineering or cost parameters;
- recommend comparable projects;
- propose imputed values and ranges;
- explain the imputation method;
- preserve the original value and record the calculation pedigree.

Imputed values should always be visibly distinguished from observed data.

Appendix B - Architecture

CRAFT consists of four main architectural layers:

1. **User Access Layer** — Enables authorized users to ask questions, submit tasks and workflows, provide information, and review or approve outputs. This layer includes web applications and interfaces for search, data ingestion, visualization, reporting and administration.
2. **Reasoning and Agent Layer** — Interprets requests, breaks complex tasks into manageable steps, retrieves relevant evidence, coordinates specialized AI agents, prepares responses and assesses confidence. It may also identify missing information and refer uncertain results for human review.
3. **Knowledge Management Layer** — Connects approved information sources and transforms dispersed information into structured, searchable and reusable organizational knowledge. This layer includes storage and retrieval services for both structured data, such as SQL databases, and unstructured information, such as documents and reports.
4. **Governance and Integration Layer** — Manages security, permissions, system connections and data movement. It applies applicable business rules, supports human review and approval, and records sources, actions and decisions to provide traceability and accountability.

Together, these layers enable CRAFT to help users locate trusted information, reason from evidence, reuse organizational knowledge and support informed decisions while maintaining appropriate security, traceability and human control.

